

Toolkit Design for Building Camera Sensor-Driven DIY Smart Homes

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Abstract

In a do-it-yourself (DIY) smart home, users can implement desired features by combining sensors and actuators. Specifically, effective sensing of home situations is crucial for building these features, and several studies have explored the use of cameras as home sensors to capture more specific scenarios. Camera sensors offer the potential to use various elements based on visual information as sensing triggers, presenting possibilities distinct from commonly used IoT sensors. These unique characteristics may necessitate new support for the DIY smart home-building process. In light of this, our doctoral research aims to (1) understand the unique user experience of camera sensors in DIY smart home building, (2) explore the potential and challenges of using cameras as sensors, and (3) propose a design framework for a toolkit that supports the camera sensor-driven DIY smart home building experience. This doctoral colloquium paper discusses our research plans, which synthesize knowledge on toolkit design to support the effective use of camera sensors in building DIY smart home features.

CCS Concepts

• **Human-centered computing** → **Empirical studies in ubiquitous and mobile computing**; **Empirical studies in interaction design**.

Keywords

DIY, Smart Home, Camera Sensor, Home Sensor, User Experience, Intelligent Home

ACM Reference Format:

Sojeong Yun and Youn-kyung Lim. 2024. Toolkit Design for Building Camera Sensor-Driven DIY Smart Homes. In *Companion of the 2024 ACM International Joint Conference on Pervasive and Ubiquitous Computing (UbiComp Companion '24)*, October 5–9, 2024, Melbourne, VIC, Australia. ACM, New York, NY, USA, 6 pages. <https://doi.org/10.1145/3675094.3678363>

1 Problem statement

1.1 Importance and Limitations of DIY Smart Homes

With advancements in sensing technologies, various actuators, and integrated connectivity between devices, the opportunities to enjoy smart home features in our daily lives are steadily increasing.



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UbiComp Companion '24, October 5–9, 2024, Melbourne, VIC, Australia
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ACM ISBN 979-8-4007-1058-2/24/10
<https://doi.org/10.1145/3675094.3678363>

Smart home companies emphasize the convenience of automation features, encouraging users to delegate more of their daily activities to smart devices. However, some studies raise concerns about the excessive use of automation within the home, suggesting that relying entirely on automation may not truly support user well-being [3, 22, 43]. Research on user experiences with these products has pointed out issues such as lack of user control [35], automation actions that are not clearly understood [47], and indiscriminate efficiency-driven automation [22], all of which can lead to negative user experiences. In this context, the concept of DIY smart homes has emerged, enabling users to combine various sensors and actuators to implement their desired smart home features [44]. Numerous DIY smart home studies have focused on helping users effectively build the features they desire. Example models include the trigger-action model [39], routine-based smart homes [45], and do-it-together smart homes [40].

Despite these efforts, DIY smart homes remain primarily adopted by enthusiast users rather than becoming mainstream. Previous studies capture various difficulties users face due to the limitations in the features that can be implemented and the complexity of building these features [15, 19, 34]. For instance, the complexity of device installation and setup can lead to accessibility differences among family members, contributing to their negative experiences [8, 23, 38]. To build DIY smart home features, users need to collect appropriate data from sensors and set rules to control the physical environment through actuators. However, learning to use sensors and actuators to build these features requires considerable time and effort [37]. Thus, one of the main goals of DIY smart homes has been to reduce the technical barriers for users in building these features [30]. A significant reason for the high entry barrier is the gap between the desired sensing situations and the sensor values, which often requires coding knowledge to bridge. Users creatively tried to use commercially available sensors such as door sensors, motion sensors, and presence sensors to sense various situations, but there are still limitations in building all the desired features [17, 25]. These limitations in sensing capabilities are one of the major obstacles preventing users from building the diverse and creative features they want.

1.2 Potential Changes in the DIY Smart Home Building Experience with the Emergence of Camera Sensors

To overcome existing limitations, more advanced sensors have recently emerged, allowing users to directly capture a variety of specific situations with a single sensor. Camera sensors are among the most representative of these advanced sensors [28, 46, 48]. By

enabling the sensing of various situations based on visual information, camera sensors can play a crucial role in realizing diverse DIY smart home needs [12, 25, 51]. However, the distinct characteristics of camera sensors, compared to commonly used IoT sensors, can change the way users build their DIY smart home features. IoT sensors commonly detect the status of objects or people's movements through numerical sensor values, while camera sensors detect situations based on visual information [28, 46, 48]. Additionally, IoT sensors typically sense specific situations by being attached to the target objects being observed, focusing on a narrow scope. In contrast, camera sensors can sense broader situations across an entire room, providing a wider scope of situational awareness [12, 25, 51]. These unique characteristics of camera sensors may necessitate different kinds of support in the DIY smart home-building process.

In light of this background, our doctoral research aims to explore the unique characteristics of camera sensors and the user experience of building DIY smart homes based on camera sensors. Ultimately, the goal is to derive a framework for designing a toolkit that can effectively support users in their camera sensor-based DIY smart home-building experience.

2 Related Work

2.1 Existing Efforts to Support Users in Building DIY Smart Homes

DIY smart home systems offer users the potential to design smart home features tailored to their specific needs. However, the increased emphasis on the user's role necessitates various types of additional support. Consequently, one of the major research areas in DIY smart homes has been developing methods to help users more effectively construct their own smart home systems [30, 37]. Many recent studies have devised methods to enable users to set up feature rules naturally and conveniently in their daily lives. These studies include approaches that proactively suggest features based on the user's regular behavior patterns [27, 53, 55] and those that facilitate the specification of feature rules through conversational interfaces [21]. Additionally, there has been significant research aimed at supporting the debugging process to ensure that DIY smart home functions operate as intended within the home [2, 6, 52, 54].

Other lines of research have explored the possibility of building DIY smart home features by detailed sensing of home situations with a single sensor, thereby reducing the complexity of rule-setting processes. Examples include audio sensors that identify the objects users are interacting with based on sound [9, 37], smart rings that determine the objects users are holding based on texture detection [42], and camera sensors that monitor user activities [28, 31, 36, 48]. Among these, camera sensors are particularly prominent due to their ability to capture comprehensive information in a single image, making them highly versatile [5, 26].

2.2 Utilizing Cameras as Home Sensors

Camera sensors operate based on visual information, enabling more intuitive and explicit interactions between the sensor and the user [12, 25, 25, 30, 37, 51]. Unlike IoT sensors with clearly defined sensing roles, camera sensors possess flexibility in their sensing roles, allowing users to define and utilize the sensors as needed [12, 25, 51]. In the field of human-computer interaction (HCI), researchers have

focused on the unique possibilities of camera sensors, studying how people use home cameras as sensors. In particular, advancements in image analysis technology allow for recognizing human activities based on the objects they are holding and their postures [28, 48], recognizing gestures [18, 31, 36], analyzing the location of objects through object tracking [46], and identifying strangers using facial recognition [14, 32]. These sensing capabilities can serve as sources for automating various smart home features. While implementing these features with commonly used IoT sensors would require complex rule settings or might even be impossible, camera sensors offer an intuitive and straightforward way to build them. Moreover, the development of Vision-Language Models (VLM) [10, 20, 29, 49] enhances the ability of camera sensors to provide detailed descriptions of the current home situation, indicating that the application of camera sensors can become even more versatile based on user definitions.

Despite the active discussions in HCI about the potential of camera sensors, research on using camera sensors to build smart home features in DIY environments is still in its early stages. Given the unique characteristics of camera sensors compared to commonly used IoT sensors, there may be a need for new types of support during the DIY smart home building process. Additionally, because camera sensors can capture detailed home situations, their use requires more delicate consideration [13, 14, 24]. Given this background, this research aims to explore the process of building camera sensor-driven DIY smart homes and to derive a framework for toolkit design that can support users in this process.

3 Research Questions and Methodology

This research focuses on the advancements in camera sensing technology and aims to design a toolkit that helps users seamlessly build DIY smart home features using camera sensors. To achieve this goal, we have planned a series of studies that progress from understanding existing experiences to exploring new possibilities. This doctoral research aims to answer four key research questions.

RQ 1. What are the characteristics of camera sensors compared to IoT sensors, and what new user experiences do they bring to the DIY smart home-building process?

In the first study, we explored the distinctive characteristics of camera sensors compared to IoT sensors and how these characteristics can introduce new user experiences in building DIY smart home features. To achieve this, we developed a DIY smart home kit using IoT sensor blocks (light, touch, proximity, motion, and tilt sensors), a camera sensor block, and actuator blocks (motor, sound, LED, and light blocks), which are designed for coding education (Figure 1) [7]. For the camera sensor, participants used a Teachable Machine [11] to train and utilize their machine-learning image classification models. Twelve participants were recruited and given seven days to freely use IoT sensors, the camera sensor, and actuators to build simulation prototypes, experiencing the DIY smart home building process (Figure 2). Through this study, we identified five key characteristics of camera sensors and analyzed how these characteristics can bring both opportunities and challenges to DIY smart home building. In summary, camera sensors can offer a new construction experience by enabling the building of features that are not

bound by typical routines, allowing for flexible modification and expansion of features even after the initial setup, encouraging consideration of the interrelationships between features, and providing the possibility for intuitive building and debugging based on visual information.

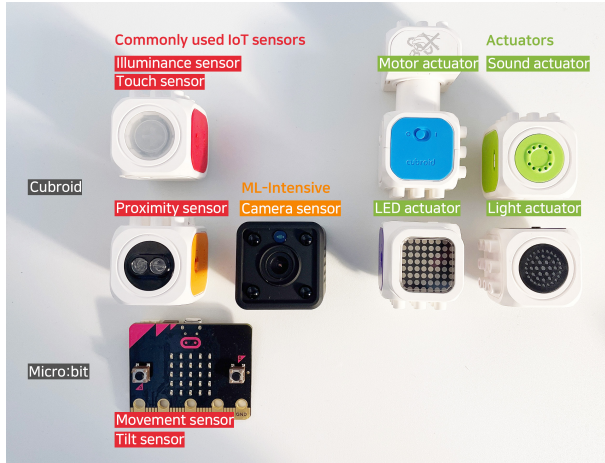


Figure 1: DIY smart home kit used in the first study [51]

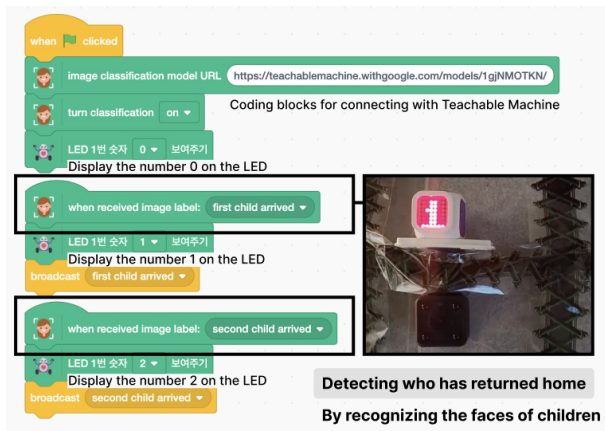


Figure 2: Example of feature built by participant [51]

RQ 2. What key functions are needed to effectively utilize camera sensors in building DIY smart homes?

In the second study, the goal was to explore what core functions camera sensors need to have for participants to build various desired features. To achieve this, we developed a technology probe [4] equipped with basic object recognition capabilities using pre-trained models (YOLO v8 [50], MediaPipe [16]), which provided the most basic functions for using a camera as a sensor. This probe acted as a sensor that could toggle sensor values on or off based on whether specified objects were present in designated areas. We hoped that by utilizing these basic functions, participants would

identify additional features that might be necessary. Along with the camera sensor probe, we provided actuators to equip participants with the basic tools needed to build a DIY smart home (Figure 3). We then conducted a two-week in-the-field diary study with 12 participants, asking participants to build various smart home features using the probe. Additionally, through interviews, we investigated how participants used the probe and their expectations for the functions that camera sensors need to have to fully realize their desired features.

The study results revealed that participants discovered various ways to effectively utilize camera sensors in a home environment. By synthesizing these expectations, we collected participants' needs for more intelligent camera sensors that go beyond merely understanding the current situation in detail. Additionally, this study explored privacy issues related to camera sensors.

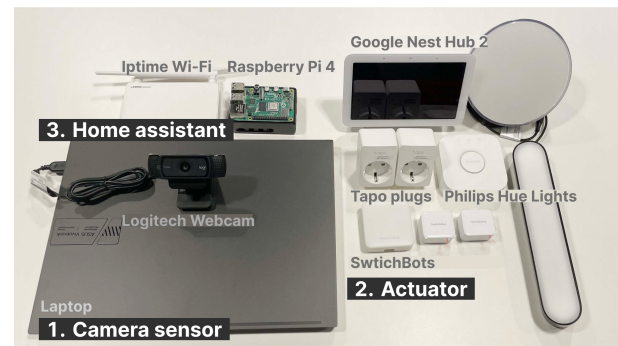


Figure 3: DIY smart home kit used in the second study

RQ 3. What features do users want to build using intelligent camera sensors, and what additional support is needed in the DIY smart home building process?

The second study revealed that users have a desire to build more intelligent DIY smart home features by contextually analyzing the current situation based on sequential, comprehensive, and everyday contexts. This possibility can be particularly realized with the advancement of VLM-based image analysis capabilities, making intelligent camera sensors a reality. These advancements mean that camera sensors can more accurately understand home situations without additional user input, and allow users to specify the focus of image analysis and the types of resulting sensor values according to their needs. Such sensors can provide a differentiated experience compared to IoT sensors and the camera sensors used in the first two studies.

Based on this background, the third study aims to explore how people want to utilize the potential of VLM-based intelligent camera sensors and what new support they need when building DIY smart homes. First, as the form of home cameras becomes more varied, we sought to classify the forms that could bring significant functional differences when using cameras as sensors. To achieve this, we proposed three concepts for camera sensor configurations based on the results of the second study and a review of various papers [1, 26, 33, 41] suggesting different forms of home cameras.

- (1) Concept 1: What if a wearable camera or home robot could follow the objects you want to observe and understand your sequential daily activities?
- (2) Concept 2: What if multiple cameras in our home could communicate with each other to comprehensively understand home situations?
- (3) Concept 3: What if a camera embedded in home appliances could analyze our daily patterns?

Under these concepts, we plan to conduct a three-week diary study with 12 participants who have experience in building smart homes, asking them to brainstorm DIY smart home ideas using intelligent camera sensors. The diary provided to participants will guide them to follow the existing DIY smart home-building process identified in the first study, allowing for a comparative analysis of the additional considerations required for a DIY smart home based on intelligent camera sensors. Through this study, we aim to explore the new potential of intelligent camera sensors and identify the support needed in the DIY smart home building process to fully realize this potential. Additionally, we aim to gather concerns experienced during the process of experiencing smart home features based on camera sensors. This study is scheduled to be conducted soon.

RQ 4. How should a toolkit be designed to support the building of camera sensor-based DIY smart homes?

To answer the fourth research question, we aim to propose a framework that includes design considerations for a toolkit to assist in building camera sensor-driven DIY smart homes, based on the results of the previous three studies. Specifically, we intend to propose a framework that considers the new values that camera sensor-based smart homes can provide to users (from the results of Studies 2 and 3) and the support needed at each phase of DIY smart home construction (from the results of Study 1) to build DIY smart home features that comprehensively encompass these values (from the results of Study 3). Additionally, we will integrate the concerns experienced by participants during Studies 2 and 3 into the framework.

4 Evaluation

After developing the framework for a toolkit to support the building of camera sensor-driven DIY smart homes, we plan to conduct a design workshop with professional designers in the smart home field. The workshop will explore and discuss the applicability and potential of the proposed design framework. During the workshop, designers will engage in activities such as redesigning the interfaces of existing DIY smart home applications, considering the proposed framework. This process will help validate the framework's usefulness and identify additional aspects that need to be considered to make the toolkit more practical.

5 (Expected) Contribution

Through this doctoral research, we anticipate making three major contributions:

- (1) Deriving knowledge on the design of cameras as home sensors: This research aims to utilize various types of camera

sensing technologies to achieve the research objectives. By collecting both the advantages and concerns of camera sensors in all planned studies, this research is expected to synthesize knowledge on designing camera sensors that maximize their potential while addressing any issues, thereby enhancing their role as home sensors.

- (2) Exploring new values in the user experience of DIY smart homes: By enabling the construction of numerous DIY smart home features based on camera sensors, this research aims to collect a variety of examples of such features. These ideas could lead to new values that differ from those typically pursued in IoT sensor-based DIY smart home systems. By gathering users' feature ideas, the research seeks to expand the knowledge of the diverse values that smart home features can bring to users.
- (3) Synthesizing knowledge on designing toolkits for building camera sensor-driven DIY smart homes: This research is expected to synthesize knowledge on the considerations necessary for designing toolkits that assist in building DIY smart homes using camera sensors, and also ensuring user control even as the sensing technology becomes more intelligent. Furthermore, as home technologies become increasingly sophisticated, this research aims to contribute to future discussions on the direction of smart home advancements and related research.

Acknowledgments

This work was supported by the National Research Foundation of Korea(NRF) grant funded by the Korea government(MSIT) (No. NRF-2021R1A2C2004263). The research is also partially funded by Asleep Inc.

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